

Continuity

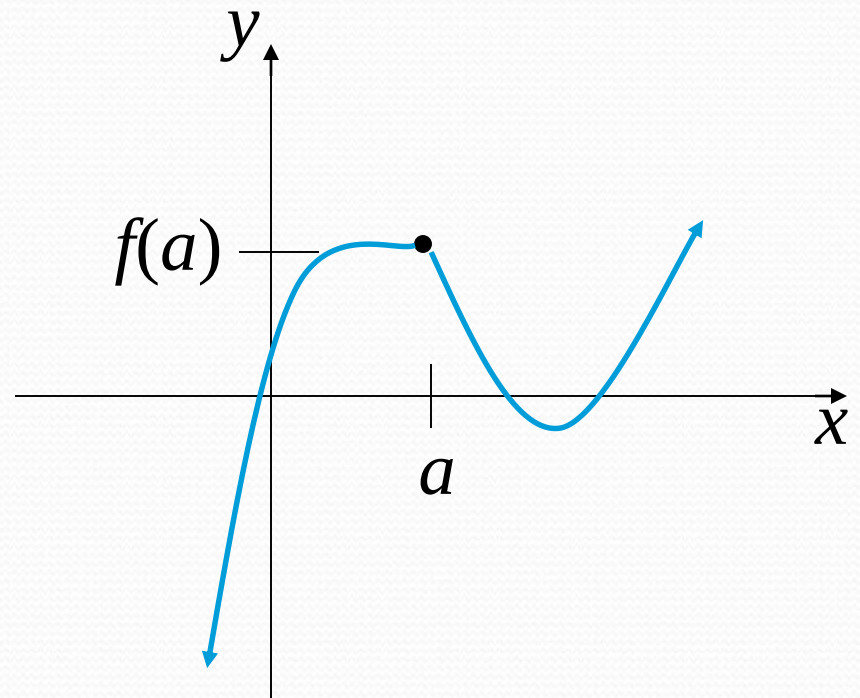
Continuity of a Function

A function f is **continuous** at the point $x = a$ if the followings are true:

i) $f(a)$ is defined

ii) $\lim_{x \rightarrow a} f(x)$ exists

iii) $\lim_{x \rightarrow a} f(x) = f(a)$



Example

Is the function f given by $f(x) = x^2 - 5$
continuous at $x = 3$? Why or why not?

1)
$$f(3) = 3^2 - 5 = 9 - 5 = 4$$

2) By the Theorem on Limits of Rational Functions,

$$\lim_{x \rightarrow 3} x^2 - 5 = 3^2 - 5 = 9 - 5 = 4$$

3) Since f is continuous at $x = 3$.

$$\lim_{x \rightarrow 3} f(x) = f(3)$$

Physical phenomena are usually continuous.

–For instance, the displacement or velocity of a vehicle varies continuously with time.

Geometrically, you can think of a function that is continuous at every number in an interval as a function whose graph has no break in it.

- The graph can be drawn without removing your pen from the paper.

Properties of Continuous Functions

The constant function $f(x)$ is continuous everywhere.

Ex. $f(x) = 10$ is continuous everywhere.

The identity function $f(x) = x$ is continuous everywhere.

Properties of Continuous Functions

If f and g are continuous at $x = a$, then

$f \pm g$, fg , and $\frac{f}{g}$ ($g(a) \neq 0$) are continuous

at $x = a$.

A polynomial function $y = P(x)$ is continuous at everywhere.

A rational function $R(x) = \frac{p(x)}{q(x)}$ is continuous
at all x values in its domain.

Problem 1**Determine whether the following function is continuous at $x=4$**

$$f(x) = \begin{cases} 2x + 3, & x \leq 4 \\ 7 + \frac{16}{x}, & x > 4 \end{cases}$$

Solution**When $x \leq 4$ then $f(x) = 2x + 3$**

Now L.H.L $= \lim_{x \rightarrow 4^-} f(x)$

$$= \lim_{h \rightarrow 0} f(4 - h)$$

$$= \lim_{h \rightarrow 0} 2(4 - h) + 3$$

$$= 8 + 3$$

$$= 11$$

When $x > 4$ then $f(x) = (7 + \frac{16}{x})$

Now R.H.L $= \lim_{x \rightarrow 4^+} f(x)$

$$= \lim_{h \rightarrow 0} f(4 + h)$$

$$= \lim_{h \rightarrow 0} \left(7 + \frac{16}{4 + h} \right)$$

$$= 7 + 4$$

$$= 11$$

Since L.H.L = R.H.L.

$$\therefore \lim_{x \rightarrow 4} f(x) = 11$$

Now, when $x = 4$ then $f(x) = 2x + 3$

$$\begin{aligned}\therefore f(4) &= 2 \times 4 + 3 \\ &= 11\end{aligned}$$

Again Since, $\lim_{x \rightarrow 4} f(x) = f(4)$

So $f(x)$ is continuous at $x = 4$

Problem 2**Solution**

If

$$f(x) = \begin{cases} -3x + 5; & -3 < x < 1 \\ 2; & 1 \leq x < 2 \\ 1 + \frac{1}{2}x; & 2 \leq x < 4 \end{cases}$$

Determine whether the function is continuous at $x=1$, $x=2$

When $x < 1$ $f(x) = -3x + 5$

$$\text{L.H.L} = \lim_{x \rightarrow 1^-} f(x)$$

$$= \lim_{h \rightarrow 0} f(1-h)$$

$$= \lim_{h \rightarrow 0} -3(1-h) + 5$$

$$= -3(1-0) + 5$$

$$= 2$$

When $x \geq 1$, then $f(x) = 2$

$$\text{R.H.L.} = \lim_{x \rightarrow 1^+} f(x)$$

$$= \lim_{h \rightarrow 0} f(1+h)$$

$$= \lim_{h \rightarrow 0} 2$$

$$= 2$$

$$\text{Since } \lim_{x \rightarrow 1^+} f(x) = 2 = \lim_{x \rightarrow 1^-} f(x)$$

$$\therefore \lim_{x \rightarrow 1} f(x) = 2$$

Now, when $x = 1$ then $f(x) = 2$

$$\therefore f(1) = 2$$

Again Since, $\lim_{x \rightarrow 1} f(x) = f(1)$

So $f(x)$ is continuous at $x = 1$

Problem 2

Solution

$$\text{If } f(x) = \begin{cases} -3x+5; & -3 < x < 1 \\ 2; & 1 \leq x < 2 \\ 1 + \frac{1}{2}x; & 2 \leq x < 4 \end{cases}$$

When $x < 2$ the $f(x) = 2$

$$\begin{aligned} \text{L.H.L} &= \lim_{x \rightarrow 2^-} f(x) \\ &= \lim_{h \rightarrow 0} f(2-h) \\ &= \lim_{h \rightarrow 0} 2 \\ &= 2 \end{aligned}$$

When $x \geq 2$, then $f(x) = 1 + \frac{1}{2}x$

$$\begin{aligned} \text{R.H.L.} &= \lim_{x \rightarrow 2^+} f(x) \\ &= \lim_{h \rightarrow 0} f(2+h) \\ &= \lim_{h \rightarrow 0} \left\{ 1 + \frac{1}{2}(2+h) \right\} \\ &= 2 \end{aligned}$$

$$\text{Since } \lim_{x \rightarrow 2^+} f(x) = 2 = \lim_{x \rightarrow 2^-} f(x)$$

$$\therefore \lim_{x \rightarrow 2} f(x) = 2$$

Now, when $x = 2$ then $f(x) = 1 + \frac{1}{2}x$

$$\begin{aligned}\therefore f(2) &= 1 + \frac{1}{2}(2) \\ &= 2\end{aligned}$$

Again Since, $\lim_{x \rightarrow 2} f(x) = f(2)$

So $f(x)$ is continuous at $x = 2$

Problem #3

Show that $f(x) = \sin x$ is continuous for every x .

Solution

$f(x) = \sin x$ and we take an arbitrary value of x say $x=a$.

Now, we have to check $\lim_{x \rightarrow a} f(x)$ exists or not.

$$\begin{aligned}\text{L.H.L.} &= \lim_{x \rightarrow a^-} f(x) = \lim_{h \rightarrow 0} f(a - h) \\ &= \lim_{h \rightarrow 0} \sin(a - h) \\ &= \lim_{h \rightarrow 0} (\sin a \cosh - \cos a \sinh) \\ &= \sin a \cdot 1 - \cos a \cdot 0 \\ &= \sin a\end{aligned}$$

$$\begin{aligned}
 \text{R.H.L.} &= \lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0} f(a + h) \\
 &= \lim_{h \rightarrow 0} \sin(a + h) \\
 &= \lim_{h \rightarrow 0} (\sin a \cosh + \cos a \sinh) \\
 &= \sin a \cdot 1 + \cos a \cdot 0 \\
 &= \sin a
 \end{aligned}$$

Since L.H.L. = $\sin a$ = R.H.L.

$$\therefore \lim_{x \rightarrow a} f(x) = \sin a \dots\dots\dots(i)$$

Now $f(x) = \sin x$

$$\therefore f(a) = \sin a \dots\dots\dots(ii)$$

We observe that from (i) and (ii) that

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Since $f(x)$ satisfies all conditions of continuity, thus the function $f(x)$ is continuous at $x = a$.

Problem # 4

A function $f(x)$ is defined as follows:

$$f(x) = \begin{cases} \cos x, & \text{if } x \geq 0 \\ -\cos x, & \text{if } x < 0 \end{cases}$$

Is $f(x)$ is continuous at $x = 0$?

Solution

When $x < 0$ then $f(x) = -\cos x$

$$\begin{aligned} \text{Now L.H.L} &= \lim_{x \rightarrow 0^-} f(x) \\ &= \lim_{h \rightarrow 0} f(0 - h) \\ &= \lim_{h \rightarrow 0} f(-h) \\ &= \lim_{h \rightarrow 0} -\cos(-h) \\ &= \lim_{h \rightarrow 0} -\cos h \\ &= -1 \end{aligned}$$

When $x \geq 0$ then $f(x) = \cos x$

$$\begin{aligned} \text{Now R.H.L} &= \lim_{x \rightarrow 0^+} f(x) \\ &= \lim_{h \rightarrow 0} f(0 + h) \\ &= \lim_{h \rightarrow 0} f(h) \\ &= \lim_{h \rightarrow 0} \cos(h) \\ &= \cos 0 \\ &= 1 \end{aligned}$$

Since $L.H.L \neq R.H.L$

So, $\lim_{x \rightarrow 0} f(x)$ does not exist.

Thus, the function $f(x)$ is not continuous at $x = 0$.

Problem # 5

Examine the continuity of the following function at $x = \frac{1}{2}$

$$f(x) = \begin{cases} x & \text{when } 0 \leq x < \frac{1}{2} \\ 1-x & \text{when } \frac{1}{2} \leq x \leq 1 \end{cases}$$

Solution

When $x < \frac{1}{2}$ then $f(x) = x$

$$\begin{aligned} \text{L.H.L} &= \lim_{x \rightarrow \frac{1}{2}^-} f(x) \\ &= \lim_{h \rightarrow 0} f\left(\frac{1}{2} - h\right) \\ &= \lim_{h \rightarrow 0} \left(\frac{1}{2} - h\right) \\ &= \frac{1}{2} - 0 = \frac{1}{2} \end{aligned}$$

Again, when $x > \frac{1}{2}$ then $f(x) = 1 - x$

$$\begin{aligned} \text{R.H.L} &= \lim_{x \rightarrow \frac{1}{2}^+} f(x) = \lim_{h \rightarrow 0} f\left(\frac{1}{2} + h\right) \\ &= \lim_{h \rightarrow 0} \left\{ 1 - \left(\frac{1}{2} + h\right) \right\} \\ &= \lim_{h \rightarrow 0} \left\{ \frac{1}{2} - h \right\} \\ &= \frac{1}{2} \end{aligned}$$

Since L.H.L = R.H.L.

$$\therefore \lim_{x \rightarrow \frac{1}{2}} f(x) = \frac{1}{2}$$

Now, when $x = \frac{1}{2}$ then $f(x) = 1 - x$

$$\therefore f\left(\frac{1}{2}\right) = 1 - \frac{1}{2} = \frac{1}{2}$$

Again Since, $\lim_{x \rightarrow \frac{1}{2}} f(x) = f\left(\frac{1}{2}\right)$

So $f(x)$ is continuous at $x = \frac{1}{2}$

Problem # 6

Compute the value for the constant of k , that will make the following function continuous at $x=1$

Solution

$$f(x) = \begin{cases} 7x - 2, & x \leq 1 \\ kx^2, & x > 1 \end{cases}$$

When $x \leq 1$ then $f(x) = 7x - 2$

$$\text{Now L.H.L} = \lim_{x \rightarrow 1^-} f(x)$$

$$= \lim_{h \rightarrow 0} f(1 - h)$$

$$= \lim_{h \rightarrow 0} \{7(1 - h) - 2\}$$

$$= 5$$

When $x > 1$ then $f(x) = kx^2$

$$\text{Now R.H.L} = \lim_{x \rightarrow 1^+} f(x)$$

$$= \lim_{h \rightarrow 0} f(1 + h)$$

$$= \lim_{h \rightarrow 0} k(1 + h)^2$$

$$= k$$

When $x = 1$ then $f(x) = 7x - 2$

$$\therefore f(1) = 7 \times 1 - 2 = 5$$

If the function is continuous then

$$\therefore \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^-} f(x) = 5$$

So, $k = 5$.

Problem # 7

Discuss the continuity of the function

$$f(x) = \begin{cases} -x; & -2 \leq x \leq 0 \\ x; & 0 < x < 1 \\ 3-x; & 1 \leq x \leq 2 \end{cases}$$

At $x = 0$ and $x = 1$.

Now, let's see how to detect discontinuities when a function is defined by a formula.

Problem # 8

Where are each of the following functions discontinuous?

$$f(x) = \frac{x^2 - x - 2}{x - 2}$$

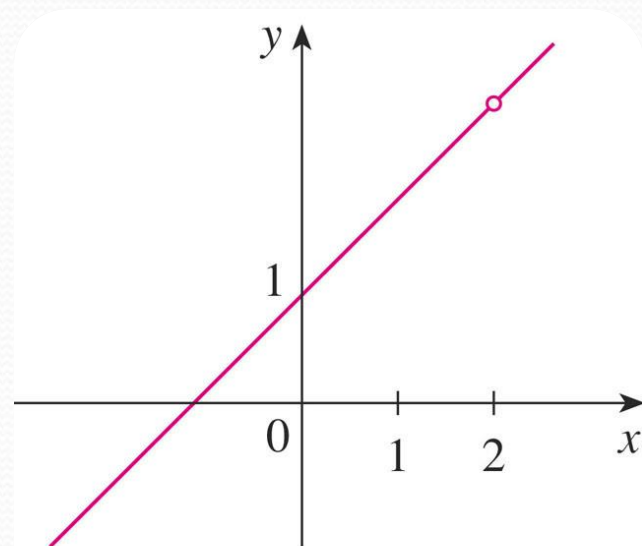
$$f(x) = \begin{cases} \frac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$$

$$f(x) = \begin{cases} \frac{x^2 - x - 2}{x - 2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$$

$$f(x) = \frac{x^2 - x - 2}{x - 2}$$

Notice that $f(2)$ is not defined.
So, f is discontinuous at 2.

The kind of discontinuity illustrated here is called removable.



$$(a) f(x) = \frac{x^2 - x - 2}{x - 2}$$

■ We could remove the discontinuity by redefining f at just the single number 2.

■ The function $g(x) = x + 1$ is continuous.

$$(g) g(x) = \frac{x - 5}{x_5 - x - 5}$$

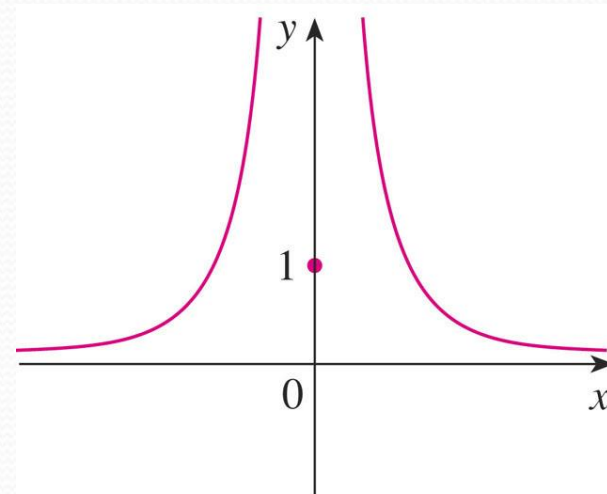
$$f(x) = \begin{cases} \frac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$$

Here, $f(0) = 1$ is defined.

However,

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{1}{x^2} \text{ does not exist.}$$

So, f is discontinuous at 0.



$$(b) f(x) = \begin{cases} \frac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$$

The discontinuity is called an infinite discontinuity.

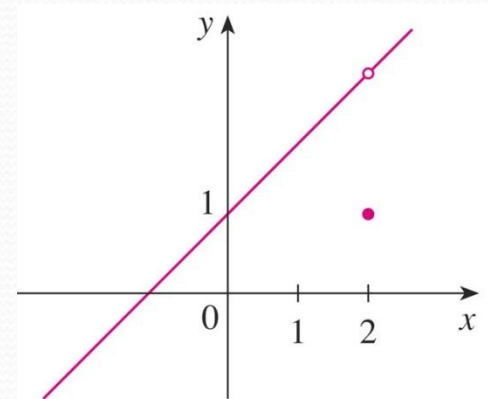
$$f(x) = \begin{cases} \frac{x^2 - x - 2}{x - 2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$$

Here, $f(2) = 1$ is defined and

$$\begin{aligned} \lim_{x \rightarrow 2} f(x) &= \lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x - 2} \\ &= \lim_{x \rightarrow 2} \frac{(x - 2)(x + 1)}{x - 2} \\ &= \lim_{x \rightarrow 2} (x + 1) = 3 \quad \text{exists.} \end{aligned}$$

$$\lim_{x \rightarrow 2} f(x) \neq f(2)$$

So, f is not continuous at 2.



$$(c) f(x) = \begin{cases} \frac{x^2 - x - 2}{x - 2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$$

The kind of discontinuity illustrated here is called removable.



Thank you